PAPER_854: LENR Neutron Production k_eta Calibration Across Three Environments with Delta-k Buoyancy Tracking

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-04 **Session:** 199 **Source:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 – Aug 07, 2025) **Calculator:** LENRKetaCalibration3EnvironmentDeltaKCalc (CP4 #438) **CVW:** v2.0.0 compliant

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Abstract

We present a UQFF-calibrated framework for the LENR neutron production constant k_eta across three distinct experimental environments: Metallic Hydride Cells (k_eta ~ 2.75e8), Exploding Wires (k_eta ~ 1.91e2), and Solar Corona (k_eta ~ 6.06e-6). The core equation eta = k_eta * exp(-[SSq]*n/26)* exp(-(pi - t)) * U_m / rho_vac unifies LENR neutron production with UQFF vacuum density dynamics. Buoyancy tracking via Delta-k residuals (Delta_k = k_expected - k_actual) reveals the U_b counterforce contribution: Delta_k_metallic ~ 7.25e8, Delta_k_wires ~ 8.09e2, Delta_k_corona ~ 3.94e-6.

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1. Core Equations

- `eta = k_eta * exp(-[SSq]*n/26) * exp(-(pi - t)) * U_m / rho_vac`
- `Delta_k = k_expected - k_actual  (buoyancy residual)`
- `Metallic Hydride: eta_obs = 1e13 cm^{-2}/s, U_m = 2.67e-31 J/m^3 => k_eta ~ 2.75e8`
- `Exploding Wires: eta_obs = 1e8 cm^{-2}/s, U_m = 3.85e-30 J/m^3 => k_eta ~ 1.91e2`
- `Solar Corona: eta_obs = 7e-3 cm^{-2}/s, U_m = 8.51e-33 J/m^3 => k_eta ~ 6.06e-6`

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2. UQFF Integration

This calculator operates as a stateless physics calculator within the CondensedPhysics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py -> bodies_*.csv data flow.

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3. Source Data

- **File:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 – Aug 07, 2025)
- **Session:** 199
- **VDS/DVP/BH:** ABSENT (general vacuum density references only)

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]}/\rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile: $\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

For this system, the local VDS sub-ratio is 0.086 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavioß connects

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 47, \quad n_{\text{channel}} = 23/26$$

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Since $p_{\mathrm{DVP}} = 47$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits its resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^-12 s** (nuclear phonon damping):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]\cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The tanh saturation envelope prevents unphysical divergence: $\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2)\cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.086	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 47$	✓ Res-

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	1.1×10^{-52} m^{-2} (UQFF vacuum term)	1.114×10^{-52} m^{-2}	Planck 2018	✓ Consistent
Proton decay rate	κ = 0.0005/day $\rightarrow$ Γ_p suppression	$< 4.17\times10^{-35}$/yr	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Srivastava, Y.N., Widom, A., Larsen, L. – Electroweak neutron production (LENR)
3. Kepler Mission DR25 – 4,034 candidates, 2,335 confirmed planets
4. Hubble Heritage Team / A. Nota (ESA/STScI) – Westerlund 2 / NGC 346 imaging
5. UQFF Calibration: kappa=0.0005/day, [SSq]=0.57, beta_i~0.603